

Pattern Explanation:

The **Composite Design Pattern** is used in the Restaurant Menu System because it helps represent the menu in a hierarchical tree structure. It allows us to treat individual items such as Burger, Pizza, and Juice the same way as grouped items such as Main Menu or Drinks Menu. A menu can contain single menu items or other submenus, and all of them are handled through the same interface.

Justification:

- It is suitable for restaurant systems because menus are naturally organized into categories and subcategories.
- It reduces complexity by avoiding separate code for each type of element.
- It makes the code cleaner and easier to maintain.
- It allows adding new menu items or sections easily without changing the existing structure.
- It makes the system more flexible, organized, and scalable for future updates.